

A Hopf-Bundle Counterexample to Hatori's Separation Condition for Commutative JB*-Triples

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Status. Candidate counterexample, likely valid, pending expert review. A compact principal circle-bundle triple $C^{\mathbb{T}}(L)$ satisfies Hatori's condition (*) exactly when the bundle has a continuous global section. The Hopf bundle therefore gives a commutative JB*-triple which fails (*).

1 Source problem

Hatori [1] associates to a complex Banach space E a set $\mathcal{Q}_E$ of extreme dual-ball functionals which expose the maximal convex subsets of the unit sphere. A set of representatives $\mathcal{P} \subset \mathcal{Q}_E$ contains one functional from every equivalence class up to unimodular scalar. His condition (*) says that some such $\mathcal{P}$ has the following property: for every $p \in \mathcal{P}$, every $\varepsilon > 0$, and every relatively weak* closed $F \subset \mathcal{P}$ with $p \notin F$, there is $a \in S(E)$ such that

$$p(a) = 1, \quad |q(a)| \leq \varepsilon \quad (q \in F). \quad (1)$$

In the Final Remarks on page 45, Hatori asks whether a commutative JB*-triple always satisfies (*).

2 Proof intuition

An abelian JB*-triple is a space $C^{\mathbb{T}}(L)$ of equivariant continuous functions on a principal circle bundle. Its relevant extreme functionals are point evaluations, and choosing one representative per maximal-face class is the same as choosing one point on every circle fiber. Thus the representative set in (*) is a set-theoretic transversal of the bundle.

Condition (*) prevents this transversal from acquiring a phase-shifted limit. Indeed, if representatives converge to λp with $\lambda \neq 1$, a relatively closed tail can be separated from p by (*); weak* continuity then says its value at λp is small, contradicting $p(a) = 1$. Hence the transversal is closed. In a compact bundle it is then a compact copy of the base, so it is a continuous global section. Conversely, a continuous section makes (*) an immediate Urysohn-lemma construction. The nontrivial Hopf bundle has no section and therefore supplies the desired counterexample.

3 Bundle characterization and counterexample

Let L be a compact principal $\mathbb{T}$ -bundle, write $X = L/\mathbb{T}$, and put

$$E = C^{\mathbb{T}}(L) := \{a \in C(L) : a(\lambda t) = \lambda a(t) \text{ for all } \lambda \in \mathbb{T}, t \in L\}.$$

7. FINAL REMARKS

In section [5](#) we introduced the condition $(*)$ for Banach spaces and we proved that an extremely C -regular closed subspace of $C_0(Y, \mathbb{K})$ satisfies the condition $(*)$. On the other hand we have not enough examples of Banach spaces which satisfy the condition. Cabezas, Cueto-Avellaneda, Hirota, Miura and Peralta [\[14\]](#) proved that every *commutative* JB^* triple satisfies the complex Mazur-Ulam property. We do not know if a commutative JB^* triple satisfies the condition $(*)$ or not. If it would satisfy the condition $(*)$, we would get an alternative proof of a theorem of Cabezas, Cueto-Avellaneda, Hirota, Miura and Peralta [\[14\]](#) [Theorem 3.1]. It is interesting to exhibit examples of Banach spaces which satisfy the condition $(*)$. We have proved the *complex* Mazur-Ulam property especially for a closed subalgebra of $C_0(Y, \mathbb{K})$ which separates the points of Y and has no common zeros, we expect it also has the Mazur-Ulam property.

Figure 1: The open question in Hatori [1, p. 45].

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With its supremum norm and pointwise triple product, E is a commutative JB^* -triple. For $t \in L$, write $\delta_t(a) = a(t)$.

We use the following standard face identification in precisely the form recorded by Cabezas, Cueto-Avellaneda, Hirota, Miura, and Peralta [2, Section 3]: the map

$$L \longrightarrow \mathcal{Q}_E, \quad t \longmapsto \delta_t, \quad (2)$$

is a weak* homeomorphism in the compact case, $\delta_{\lambda t} = \lambda \delta_t$, and the equivalence classes defining a representative set are exactly the circle orbits. Equivalently, a set of representatives is a subset of L meeting every fiber exactly once.

Theorem 1. *For a compact principal circle bundle $\pi : L \rightarrow X$, the commutative JB^* -triple $C^\mathbb{T}(L)$ satisfies Hatori's condition $(*)$ if and only if π admits a continuous global section. Equivalently, $(*)$ holds if and only if the principal bundle is trivial.*

Proof. Suppose first that $(*)$ holds, witnessed by a representative set $\mathcal{P} \subset \mathcal{Q}_E$. We claim that $\mathcal{P}$ is weak* closed in $\mathcal{Q}_E$. Let $r \in \overline{\mathcal{P}}$. By the representative property and (2), there are unique $p \in \mathcal{P}$ and $\lambda \in \mathbb{T}$ such that $r = \lambda p$.

Assume $\lambda \neq 1$. The compact Hausdorff space $\mathcal{Q}_E$ has an open neighborhood U of r whose closure does not contain p . Set

$$F = \mathcal{P} \cap \overline{U}.$$

Then F is relatively closed in $\mathcal{P}$, $p \notin F$, and $r \in \overline{F}$: every neighborhood of r meets $\mathcal{P} \cap U$ because $r \in \overline{\mathcal{P}}$. Apply $(*)$ with $\varepsilon = 1/2$. There is $a \in S(E)$ with

$$p(a) = 1, \quad |q(a)| \leq \frac{1}{2} \quad (q \in F).$$

The function $q \mapsto q(a)$ is weak* continuous. Since $r \in \overline{F}$, it follows that $|r(a)| \leq 1/2$. On the other hand, $r = \lambda p$ gives $|r(a)| = |\lambda p(a)| = 1$, a contradiction. Hence $\lambda = 1$, so $r = p \in \mathcal{P}$. This proves that $\mathcal{P}$ is closed.

Via (2), regard $\mathcal{P}$ as a closed transversal in L . It is compact and the restriction

$$\pi|_{\mathcal{P}} : \mathcal{P} \longrightarrow X$$

is a continuous bijection. Since X is Hausdorff, this is a homeomorphism. Its inverse is a continuous global section of π .

Conversely, let $s : X \rightarrow L$ be a continuous section. It trivializes the bundle by the homeomorphism

$$X \times \mathbb{T} \longrightarrow L, \quad (x, \lambda) \longmapsto \lambda s(x).$$

Choose the representatives $\mathcal{P} = \{\delta_{s(x)} : x \in X\}$. Let $p = \delta_{s(x_0)} \in \mathcal{P}$, and let $F \subset \mathcal{P}$ be relatively closed with $p \notin F$. Under the homeomorphism $X \cong \mathcal{P}$, the set F corresponds to a closed set $C \subset X$ not containing x_0 . By Urysohn's lemma there is a continuous $h : X \rightarrow [0, 1]$ such that $h(x_0) = 1$ and $h|_C = 0$. Define

$$a(\lambda s(x)) = \lambda h(x).$$

Then $a \in S(C^{\mathbb{T}}(L))$, $p(a) = 1$, and $q(a) = 0$ for every $q \in F$. Thus (1) holds for every $\varepsilon > 0$. $\square$

Theorem 2. *Let $\mathbb{T}$ act on*

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}$$

by scalar multiplication. The commutative JB^ -triple*

$$C^{\mathbb{T}}(S^3) = \{a \in C(S^3) : a(\lambda z) = \lambda a(z)\}$$

does not satisfy Hatori's condition ().*

Proof. The quotient map $S^3 \rightarrow S^3/\mathbb{T} \cong S^2$ is the Hopf principal circle bundle. It has no continuous global section. Indeed, a section would trivialize the principal bundle and give a homeomorphism $S^3 \cong S^2 \times S^1$, which is impossible because

$$\pi_1(S^3) = 0, \quad \pi_1(S^2 \times S^1) \cong \mathbb{Z}$$

(see, for example, [3]). The conclusion follows from Theorem 1. $\square$

4 Scope, verification, and novelty

Theorem 2 answers Hatori's question negatively. The stronger Theorem 1 identifies the exact obstruction for all compact principal-circle-bundle models: condition (*) detects bundle triviality. This does not contradict the complex Mazur–Ulam theorem of Cabezas et al.; in fact $C^{\mathbb{T}}(S^3)$ still has the complex Mazur–Ulam property [2, Theorem 3.1]. It shows that Hatori's sufficient condition is not necessary, even within commutative JB^* -triples.

No computation is used. The proof audit in `VERIFICATION.md` isolates the face-representation input, the closed-transversal argument, and the Hopf obstruction.

The local run indexes were searched for arXiv:2205.01888, the exact title, Mazur–Ulam/Tingley terms, and condition (*). Bounded web searches on August 9, 2026 used the exact question, title and author, “commutative JB^* -triple,” “condition (*),” principal circle bundles, and the Hopf fibration.

They located the source and the known complex Mazur–Ulam theorem but no later answer or this bundle characterization. Novelty confidence is positive but conservative pending expert literature review.

Human review is strongly recommended. The decisive point is that $(*)$ forces an arbitrary representative transversal to be closed; once that is confirmed, the compact-bundle topology makes the counterexample immediate.

References

- [1] O. Hatori, *The Mazur–Ulam property for a Banach space which satisfies a separation condition*, RIMS Kôkyûroku Bessatsu **B93** (2023), 29–82; arXiv:2205.01888.
- [2] D. Cabezas, M. Cueto-Avellaneda, D. Hirota, T. Miura, and A. M. Peralta, *Every commutative JB^* -triple satisfies the complex Mazur–Ulam property*, Ann. Funct. Anal. **13** (2022), Article 60; arXiv:2201.06307; doi:10.1007/s43034-022-00204-6.
- [3] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002.